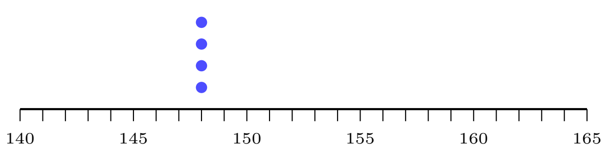
Statistical Questions and How Spread Out the Data Is

Deciding which questions are statistical, then describing a distribution by its centre and by how spread out it is

How to use this sheet.

- A question is *statistical* when you expect the answers to *vary*. The test on this sheet is concrete: go and collect the answers, then look at the dot plot. All the dots in one stack means no variability, and no variability means the question was not statistical.
- Every data set is drawn as a dot plot on a number line. One dot is one answer, and dots stack up when two people gave the same answer.
- Two numbers describe a distribution, and you need both: a **centre** (mean or median) and a **spread** (range or interquartile range). Reporting the centre alone is the mistake problem 10 is about.
- Quartiles here are the school method: sort the values, split them into a lower half and an upper half, and take the median of each half. IQR $= Q_3 - Q_1$.

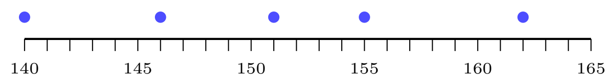
1. **Not every question is statistical.** Priya asks four classmates: "How tall is Priya?" Here are the four answers she collected, in centimetres.



Largest answer _________ − smallest answer _________ = range _________

Do the answers vary? __________ So is "How tall is Priya?" a statistical question? __________

2. Now change one word. Priya asks five classmates: "How tall are the students in our class?" Here are their heights in centimetres.



Largest _________ − smallest _________ = range _________

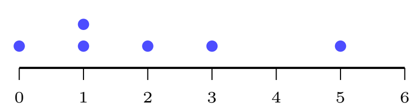
Do the answers vary? __________ Is this a statistical question? __________

What single word made the difference between problem 1 and problem 2? _________________________

3. Sort the four questions. For each question below, somebody has already gone and collected the answers. Compute the range of each answer set, then tick the questions that are statistical.

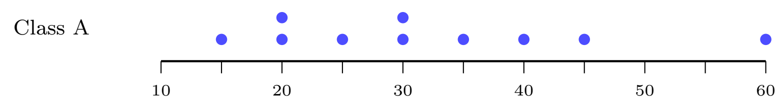
question	answers collected	range	statistical?
(a) How old is our teacher?	38, 38, 38	_________	
(b) How old are the students in Grade 6?	11, 11, 12, 12, 13	_________	
(c) How many pets does Sam have?	2, 2, 2	_________	
(d) How many pets do the students in our class have?	0, 1, 1, 2, 3, 5	_________	

Question (d) drawn as a dot plot:



Write the rule you used in your own words: ____

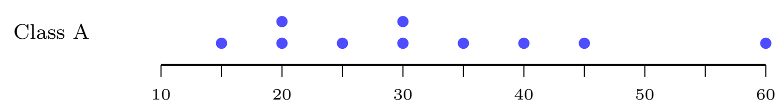
4. Class A: find the centre. Ten students in Class A recorded how many minutes they spent on homework last night.



Add all ten values: _________

Divide by how many values there are: mean = _________minutes

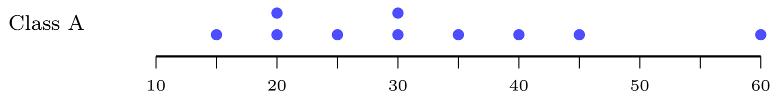
5. Class A: find the range. Same data as problem 4.



Largest value _________ Smallest value _________ Range _________minutes

In one sentence, what does that range tell a teacher that the mean does not? ________________________________

6. Class A: find the median — carefully. Sam looks at the unsorted list $30, 45, 20, 15, 60, 30, 25, 40, 20, 35$, picks the value in the middle of the list as written, and reports a median of 35 minutes.

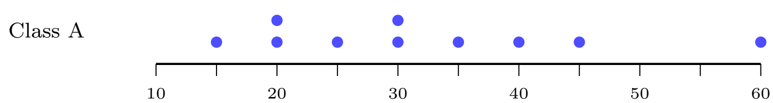


Sort the values first (the dot plot has already done this for you). With ten values, the median is halfway between the 5th and the 6th.

5th value ________ 6th value ________ median ________minutes

What exactly did Sam do wrong? ________________________________

7. Class A: the middle half. The range uses only the two most extreme values, so one unusual student can control it. The *interquartile range* looks at the middle half instead.

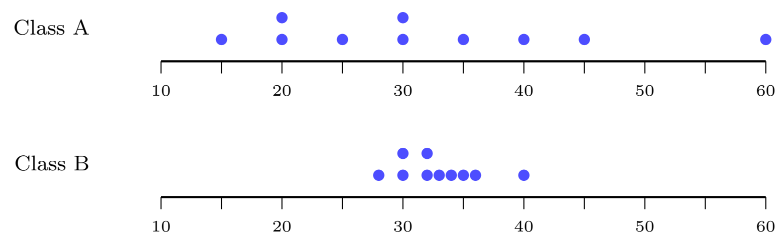


Split the ten sorted values into a lower half of five and an upper half of five.

Q_1 = median of the lower half = ________ Q_3 = median of the upper half = ________

IQR = $Q_3 - Q_1$ = ________minutes

8. Class B, and a comparison. Class B did the same homework survey. Both dot plots use the same scale, so you can compare them by eye before you compute anything.



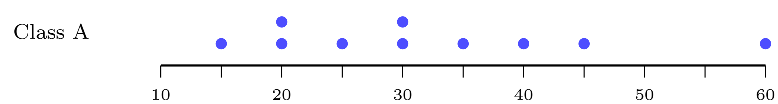
Class B: largest _________ smallest _________ range _________

Class B: $Q_1 =$ _________ $Q_3 =$ _________ IQR = _________

Class A's range was 45. How much wider is it than Class B's? _________minutes

Which class is easier to predict, and why? ________________________________

9. One student picked at random. A name is drawn at random from Class A's ten students.

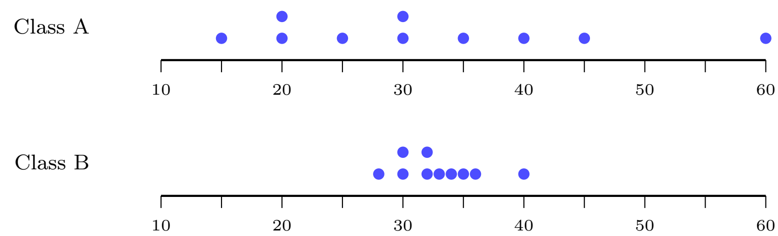


Count the dots strictly to the right of the median, 30 minutes.

Favourable outcomes _________ Total outcomes _________

P(more than 30 minutes) = _________ (write it as a fraction in lowest terms)

10. Write to the teacher. Class A's mean is 32 minutes and Class B's mean is 33 minutes — almost the same number.



Using the spreads you computed in problems 7 and 8, explain in three or four sentences why reporting only the mean would give the teacher a misleading picture of these two classes, and say what you would report instead.